Towards a portable multimodal laboratory for gait and physiological assessment in neurological gait disorders

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Introduction

Objective gait assessment is important for quantifying motor impairments and monitoring recovery in individuals with neurological disorders [1]. However, instrumented systems in rehabilitation clinics are generally expensive, non-portable, and time-consuming; therefore, clinicians usually rely on subjective clinical scales based on visual observation. Wearable-based gait analysis overcomes these limitations, providing objective data with the potential to extend monitoring into the home setting [2] [3].

Despite these advantages, integrating multiple sensor modalities remains challenging: devices such as IMUs, EMG, and pressure insoles are typically produced by different manufacturers, which complicates synchronization without extensive post-processing. This is further compounded by their reliance on wireless transmission, which can introduce connection errors and delays [4].

In addition to biomechanical measurements, physiological monitoring is essential for a more comprehensive assessment of patient performance, which may be influenced by factors such as fatigue, motivation, and overall health status [5] [6].

To address these challenges, this project aims to develop a portable, easy-to-use multimodal system that integrates synchronized biomechanical and physiological measurements. As a first step, this study evaluates the synchronization performance of the biomechanical sensors integrated into the system. Following clinical validation, the long-term goal is to enable comprehensive gait assessment beyond the laboratory, including the home environment.

Methods

The proposed system will integrate EMG and IMU sensors (Delsys, USA), plantar pressure insoles (Moticon, Germany), and physiological sensors, synchronized via Lab Streaming Layer (LSL) through a clinician-oriented interface.

Preliminary synchronization between the Delsys and Moticon sensors was assessed using a foot-tap protocol, repeated every 10 seconds for 5 minutes across 10 trials, generating simultaneous acceleration peaks in both sensors. The resultant acceleration magnitude was computed per sensor, and cross-correlation analysis was applied to estimate the inter-sensor delay for each event.

Results

Across the 10 trials, each represented by a different color in Figure 1, the overall mean synchronization delay was approximately −32 ± 24 ms. In seven trials, the delay fluctuated around a stable mean, with regression slopes below 0.02 ms/s, indicating no systematic drift.

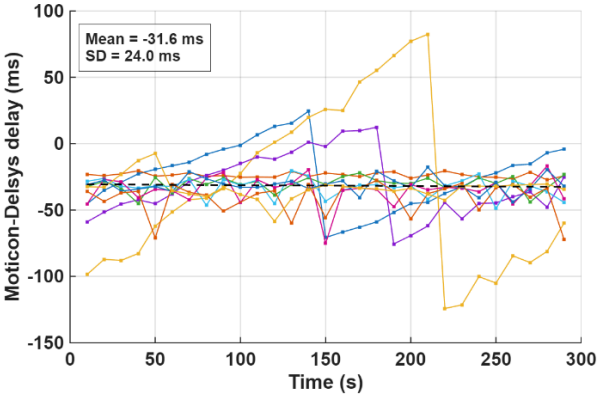


Figure 1: Delay between Delsys and Moticon sensors over time across synchronization test repetitions.

In the remaining three trials, the delay increased progressively, with markedly higher slopes, indicating pronounced drift. An abrupt shift was observed approximately halfway through the recordings, after which the delay continued to increase at a similar rate.

Conclusions

Preliminary findings support the feasibility of the proposed system, with a synchronization delay below 0.1 s in 90% of trials, meeting the design requirements established for the device. However, further investigation is needed to address the drift observed in some trials. Future work will focus on further gait validation trials and on synchronizing the physiological sensors, in order to fully validate the device.

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